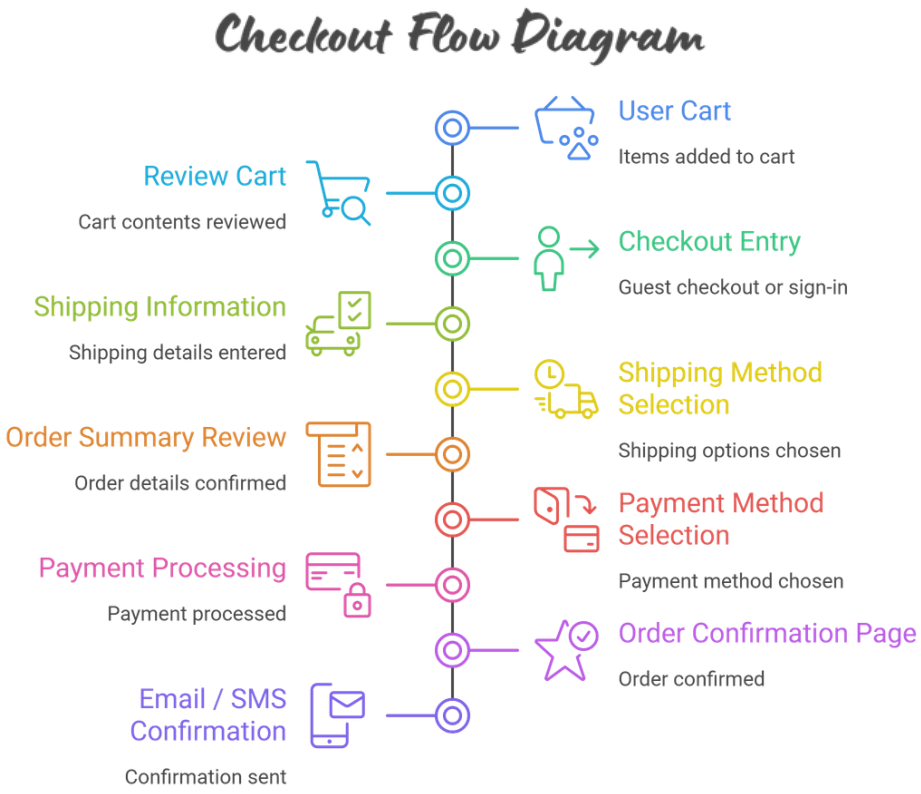
WEEKLY INTERNSHIP ASSIGNMENT 7



Checkout Stage	What Happens	UX Improvement / Purpose
User Cart	Users add products to their shopping cart and see item details and subtotal.	Show estimated shipping cost and delivery time early to reduce uncertainty.
Review Cart	Users review items, adjust quantities, or remove products before checkout.	Clear price breakdown and visible order summary help users confirm purchases easily.
Checkout Entry (Guest / Sign-in)	Users choose to continue as a guest, sign in, or create an account.	Guest checkout reduces friction and prevents abandonment caused by forced account creation.
Shipping Information	Users enter delivery details such as name, address, email, and phone number.	Inline validation, autofill suggestions, and clear error messages reduce mistakes and cognitive load.
Shipping Method Selection	Users choose a delivery option (e.g., standard or express shipping).	Display shipping cost and estimated delivery date clearly before payment to improve transparency.
Order Summary Review	Users review final order details including items, shipping, taxes, and total cost.	Persistent order summary allows users to verify all information before paying.
Payment Method Selection	Users select a preferred payment option such as card or digital wallet.	Grouped payment methods with clear icons simplify selection and improve usability.

Payment Processing	The system securely processes the payment transaction.	Loading indicators, clear error messages, and retry options maintain trust and reduce confusion.
Order Confirmation Page	Users receive confirmation that their order has been successfully placed.	Displays order number, details, and delivery estimate to reassure the user.
Email / SMS Confirmation	Confirmation message is sent to the user after checkout.	Provides receipt, order summary, and tracking information to keep users informed.

Friction Reduction Report

The checkout process is one of the most critical stages in an e-commerce journey. Even when users successfully add products to their cart, many abandon the process before completing payment. Research by the Baymard Institute shows that checkout friction is one of the primary causes of cart abandonment.

This report audits a typical checkout experience and proposes improvements focused on:

- reducing cognitive load
- improving payment transparency
- improving delivery clarity
- simplifying the checkout flow

The goal is to design a checkout experience that increases user confidence and improves completion rates.

Problem Statement

Many e-commerce checkout systems suffer from usability issues that create friction during the purchase process.

Common issues include:

- hidden delivery charges
- forced account creation
- complex payment interfaces
- lack of clear delivery estimates
- poor error messaging

These problems increase uncertainty and cause users to abandon the checkout process.

This report evaluates an existing checkout flow and proposes UX improvements to reduce friction and improve conversion.

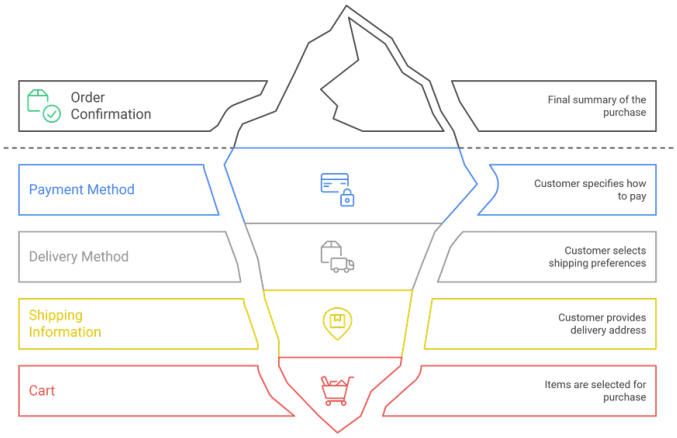
Methodologies

Two primary UX research approaches were used for the evaluation.

Funnel Analysis

Funnel analysis examines how users move through the checkout process and identifies where users drop off.

Typical checkout funnel stages: (Bottom to Top)



Each stage represents a potential drop-off point. If users encounter difficulty or confusion, they may abandon the checkout process.

Funnel analysis helps identify which stage creates the most friction.

Error State Mapping

Error state mapping analyzes situations where users make mistakes or encounter system errors.

Error Scenario	Problem	UX Improvement
Invalid address	unclear validation	inline error messages
Payment failure	vague error message	explain issue and allow retry
Missing required fields	unclear field indicators	highlight required fields
Incorrect card number	delayed validation	real-time validation

Usability Evaluation

The checkout flow was evaluated using the usability principles proposed by Jakob Nielsen.
The most relevant heuristics include:

- Visibility of system status
- Error prevention
- Recognition rather than recall
- User control and freedom
- Consistency and standards

Several issues were identified during the evaluation.

Identified Friction Points

1. Forced Account Creation

Many checkout systems require users to create an account before purchasing. This introduces unnecessary friction, especially for first-time customers.
Impact:

- increases checkout abandonment
- adds extra steps before payment

2. Hidden Shipping Costs

Shipping costs are sometimes displayed only at the final step of checkout.

Impact:
- users feel misled
- sudden price increase reduces trust

3. Complex Payment Layout
Multiple payment methods are often presented without clear organization.
Impact:
- increases cognitive load
- users struggle to identify the preferred payment option

4. Poor Delivery Transparency
Users often do not see delivery estimates until late in the checkout process.
Impact:
- reduces purchase confidence
- increases uncertainty

5. Lack of Order Summary Visibility
When users cannot easily see the order summary while entering checkout information, they may lose track of pricing details.
Impact:
- reduced trust
- increased hesitation before payment

Proposed UX Improvements
Based on the evaluation, the following improvements are recommended.
6.1 Guest Checkout Option
Allow users to complete purchases without creating an account.
Benefits:
- reduces friction for new customers
- speeds up checkout process
- improves conversion rates

Users can optionally create an account after the purchase.

6.2 Early Shipping Cost Display
Shipping costs and delivery estimates should be displayed early in the checkout process.
Example:
Standard Delivery – FREE – 4–5 days
Express Delivery – ₹99 – 1–2 days
Benefits:
- improves transparency
- prevents pricing surprises

6.3 Simplified Checkout Structure

1. Shipping Address
2. Delivery Options
3. Payment Method
4. Order Confirmation

Benefits:

- reduces cognitive load
- helps users understand progress

6.4 Persistent Order Summary

Display the order summary throughout the checkout process.

Example information:

- product list
- subtotal
- shipping cost
- taxes
- total amount

Benefits:

- improves transparency
- helps users maintain awareness of pricing

6.5 Organized Payment Method Layout

Group payment options into clear categories:

- Credit / Debit Card
- UPI
- Wallet
- Cash on Delivery

Benefits:

- improves readability
- reduces confusion during payment selection

Cognitive Load Reduction

Cognitive load refers to the mental effort required to complete a task.

Checkout interfaces should minimize this effort by:

- reducing the number of form fields
- grouping related information
- displaying clear labels and instructions
- using recognizable payment icons

These changes make the checkout process easier and faster.

A/B Testing Proposal

An A/B test can evaluate whether UX improvements increase checkout completion rates.

Test Example: Guest Checkout vs Account Creation

Variant A: Account creation required before checkout	Variant B: Guest checkout allowed

Metrics measured:

- checkout completion rate
- cart abandonment rate
- time to complete checkout

Expected outcome: Guest checkout improves conversion.

Conclusion

Checkout friction significantly impacts e-commerce conversion rates. Small usability issues can cause users to abandon the purchasing process even when they intend to buy.

By improving:

- checkout structure
- payment layout
- delivery transparency
- error handling

Platforms can create a smoother and more trustworthy purchasing experience.

A well-designed checkout system not only improves usability but also strengthens customer confidence and long-term retention.

Optimized Wireframe (HTML instead of .fig)

Github Pages > Runarok > GenAI-plus > Coursework > ux-game-intern > Assignment 07